

Step 6: Set Up Custom Domain and SSL Certificate

Deploy Spring Boot Apps to AWS Lightsail

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1 Welcome to Step 6

Configure a custom domain and SSL certificate for your Spring Boot application

What we'll accomplish: - Set up professional domain name - Enable HTTPS encryption - Configure automatic certificate renewal - Ensure production-ready security

This final step transforms your application from a development prototype into a production-ready service that users can trust and easily remember.

2 Why SSL and Custom Domains Matter

SSL Benefits: - Encrypts data transmission - Browser trust indicators - SEO ranking improvements - Protection from attacks

Custom Domain Benefits: - Professional appearance - Easy to remember URLs - Brand recognition - User confidence

Modern browsers actively warn users about non-HTTPS sites, making SSL essential. Custom domains provide credibility and make your application memorable for users.

3 Prerequisites Check

Before we begin, ensure you have:

- Registered domain name
- Access to DNS management
- Lightsail instance running
- Nginx configured (from Step 5)
- Spring Boot app accessible via IP

If any of these prerequisites are missing, go back and complete the previous steps before proceeding with domain and SSL configuration.

4 Setting Up Lightsail DNS Zone

Step 1: Create DNS Zone 1. Navigate to **Networking** → **DNS zones** 2. Click **Create DNS zone** 3. Enter your domain name 4. Click **Create DNS zone**



Tip

The DNS zone centralizes all DNS record management for your domain

Lightsail's DNS zones provide a simple interface for managing DNS records without needing to work directly with Route 53 or external DNS providers.

5 Configuring DNS Records

Required DNS Records:

```
# A Record: Points domain to your instance
example.com → Your Lightsail Instance IP

# CNAME Record: Points www to root domain
www.example.com → example.com
```

Configuration Steps: - **Subdomain:** Leave blank for root, enter **www** for www - **Resolves to:** Select your Lightsail instance

The A record creates the primary connection between your domain and server, while the CNAME record ensures both www and non-www versions work correctly.

6 Update Domain Registrar

Steps: 1. Copy Lightsail name servers 2. Access registrar control panel 3. Replace existing name servers 4. Save changes

Example Name Servers:

ns-123.awsdns-12.com
ns-456.awsdns-45.net
ns-789.awsdns-78.org
ns-012.awsdns-01.co.uk

DNS propagation takes up to 48 hours

This step tells the internet where to find your domain. The name servers act like a phone book, directing traffic to your Lightsail DNS zone.

7 Installing SSL Tools

SSH into your instance and install Certbot:

```
# Update package list
sudo apt update

# Install Certbot and Nginx plugin
sudo apt install certbot python3-certbot-nginx -y

# Verify installation
certbot --version
```

Certbot is the standard tool for obtaining and managing Let's Encrypt SSL certificates. The Nginx plugin automates the configuration process.

8 Configure Nginx for Domain

Update your Nginx configuration:

```
sudo nano /etc/nginx/sites-available/default
```

```
server {
    listen 80;
    server_name example.com www.example.com;

    location / {
        proxy_pass http://localhost:8080;
        proxy_set_header Host $host;
        proxy_set_header X-Real-IP $remote_addr;
        proxy_set_header X-Forwarded-For $proxy_add_x_forwarded_for;
        proxy_set_header X-Forwarded-Proto $scheme;
    }
}
```

The `server_name` directive tells Nginx which domains this configuration should handle. The proxy headers ensure your Spring Boot app receives correct client information.

9 Test and Reload Nginx

```
# Test configuration syntax
sudo nginx -t

# Reload Nginx with new configuration
sudo systemctl reload nginx
```

Expected output:

nginx: configuration file /etc/nginx/nginx.conf test is successful

Always test Nginx configuration before reloading. A syntax error could cause Nginx to fail, making your site inaccessible.

10 Obtain SSL Certificate

Use Certbot to automatically get and install SSL:

```
sudo certbot --nginx -d example.com -d www.example.com
```

Certbot will prompt for: {incremental} - Email address for notifications - Agreement to terms of service - Optional email sharing with EFF - Automatic HTTPS redirect (recommended: Yes)

Certbot handles the entire SSL certificate process: domain validation, certificate generation, Nginx configuration updates, and HTTPS redirect setup.

11 Automatic Certificate Renewal

SSL certificates expire every 90 days. Set up automatic renewal:

```
# Test renewal process (dry run)
sudo certbot renew --dry-run

# Check renewal timer status
sudo systemctl status certbot.timer

# Enable timer if needed
sudo systemctl enable certbot.timer
sudo systemctl start certbot.timer
```

Let's Encrypt certificates are short-lived for security. The automatic renewal system ensures your certificates stay valid without manual intervention.

12 Update Spring Boot for HTTPS

Production Properties:

```
# application-prod.properties
server.forward-headers-strategy=native
server.tomcat.use-relative-redirects=true
server.servlet.session.cookie.secure=true
server.servlet.session.cookie.http-only=true
```

These settings ensure your app works correctly behind the Nginx HTTPS proxy.

The forward-headers-strategy tells Spring Boot to trust headers from Nginx, while cookie settings enhance security for HTTPS connections.

13 Security Configuration (Optional)

If using Spring Security, add HTTPS enforcement:

```
@Configuration
@EnableWebSecurity
public class SecurityConfig {
    @Bean
    public SecurityFilterChain filterChain(HttpSecurity http) throws Exception {
        http
            .requiresChannel(channel ->
                channel.requestMatchers(r -> r.getHeader("X-Forwarded-Proto") != null)
                    .requiresSecure()
            )
            .headers(headers ->
                headers.httpStrictTransportSecurity(hstsConfig ->
                    hstsConfig.maxAgeInSeconds(31536000)
                        .includeSubdomains(true)))
            ;
        return http.build();
    }
}
```

This configuration adds HTTP Strict Transport Security headers and ensures all requests use HTTPS, providing additional security layers.

14 Rebuild and Deploy

Deploy your updated application:

```
# Build locally
./mvnw clean package -DskipTests

# Transfer to server
scp target/myapp.jar ubuntu@your-domain.com:~/

# Deploy on server
sudo systemctl stop myapp
sudo cp /home/ubuntu/myapp.jar /opt/myapp/
sudo systemctl start myapp
```

This deployment cycle should become familiar. Always stop the service before replacing the JAR file to ensure a clean restart with new configurations.

15 Verify Your HTTPS Setup

Testing checklist: {incremental} 1. Navigate to <https://yourdomain.com> 2. Verify SSL certificate (lock icon) 3. Confirm HTTP → HTTPS redirect 4. Test with SSL Labs: ssllabs.com/ssltest

```
# Command line verification
curl -I https://yourdomain.com
```

Thorough testing ensures your users will have a seamless, secure experience. The SSL Labs test provides a comprehensive security analysis.

16 Troubleshooting Common Issues

Domain not resolving:

```
dig yourdomain.com
nslookup yourdomain.com
```

SSL certificate problems:

```
curl -I http://yourdomain.com
sudo systemctl status nginx
sudo tail -f /var/log/nginx/error.log
```

Mixed content warnings: - Update hardcoded HTTP URLs - Use protocol-relative URLs (`//example.com/resource`)

Most issues stem from DNS propagation delays or firewall configurations. These diagnostic commands help identify the root cause quickly.

17 Certificate Management

Monitor certificate status:

```
# View all certificates
sudo certbot certificates

# Check certificate details
openssl x509 -in /etc/letsencrypt/live/yourdomain.com/cert.pem -text -noout

# Manual renewal (if needed)
sudo certbot renew
```

Regular monitoring ensures you catch any renewal issues before certificates expire. The automatic system is reliable, but manual oversight provides peace of mind.

18 Summary: What We Accomplished

Production-Ready Features: - Custom domain with professional appearance - SSL certificate with HTTPS encryption - Automatic certificate renewal - Security headers and browser trust - SEO-friendly HTTPS configuration

Security Enhancements: - Encrypted data transmission - Protection from man-in-the-middle attacks - Browser security indicators - Enhanced user confidence

Your application now meets enterprise standards for security and professionalism. Users can trust and easily access your Spring Boot application.

19 Next Steps and Best Practices

Recommended next actions: - Set up monitoring for domain/certificate health - Implement additional security headers - Plan for scaling beyond Lightsail if needed - Regular security audits and updates

Maintenance schedule: - Monthly SSL certificate checks - Quarterly security reviews - Monitor application performance and logs

With SSL and custom domain configured, your focus shifts to monitoring, maintenance, and potential scaling as your application grows in usage.

20 Congratulations!

Your Spring Boot application is now: - **Accessible** via custom domain - **Secure** with HTTPS encryption

- **Professional** with browser trust indicators - **Production-ready** for real users

You've successfully deployed a complete Spring Boot application to AWS Lightsail!

This represents a significant achievement - you've taken an application from development through to production deployment with enterprise-grade security and accessibility.